

(PART A – 2 Marks)

1. State the functions of the TCP/IP protocol.

- Segmentation and Reassembly.
- Reliable Data Transmission.
- Addressing and Routing.
- Error Detection.
- Flow Control and Congestion Control.

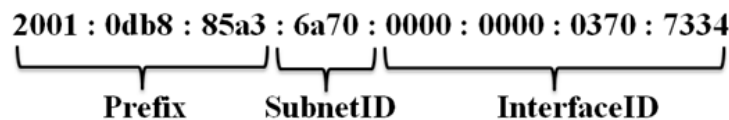
2. Enumerate the major components of DNS.

The Domain Name System (DNS) turns domain names into IP addresses, which browsers use to load internet pages. The major components are,

- Namespace
- Name servers
- Resolvers

3. For the IPv6 address find the prefix, subnet ID and Interface ID.

2001 : 0db8 : 85a3 : 6a70 : 0 : 0 : 0370 : 7334



- Prefix
 - o The prefix represents the network portion of the IP address.
- Subnet ID
 - o The subnet ID is a portion of the address used for subnetting within a larger network.
- Interface ID
 - o The Interface ID identifies a specific device on a network.

4. State two ways HTTPS improves web security.

- **Data Encryption** – HTTPS encrypts the communication between the user's browser and the web server using SSL.
- **Authentication & Integrity** – HTTPS ensures that users are connected to the legitimate website and prevents modification of data during transmission.

5. Give any four examples for a webserver.

- Internet Information Server
- Apache HTTP Server
- Nginx Web Server
- LiteSpeed Web Server

6. Why is SSH preferred for remote administration of servers?

- Encryption
- Authentication
- Integrity Verification
- Secure Key Exchange
- Tunneling & Port Forwarding

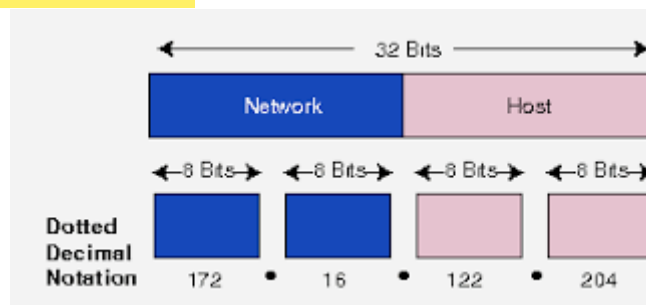
(PART B – 13 Marks)

1. Describe the classification of IP addresses into various classes and highlight the key features of each.

An IP address is the identifier that enables your device to send or receive data packets across the internet. It holds information related to your location and therefore making devices available for two-way communication.

IP address format

The 32-bit IP address is grouped eight bits at a time, separated by dots and represented in decimal format. This is known as dotted decimal notation as shown in fig. Each bit in the octet has a binary weight. The minimum value for an octet is 0, and the maximum value for an octet is 255.



Public IP Address

The public IP address is the unique address given to all computers attached to the network. The user has no control over the Public IP address as it is provided to him by the ISP whenever the machine connected to the internet.

Private IP Address

Private addresses are the addresses that are used by private networks like home or office networks. Here the logic is that these addresses are used within a single administration and not on the global network or the internet.

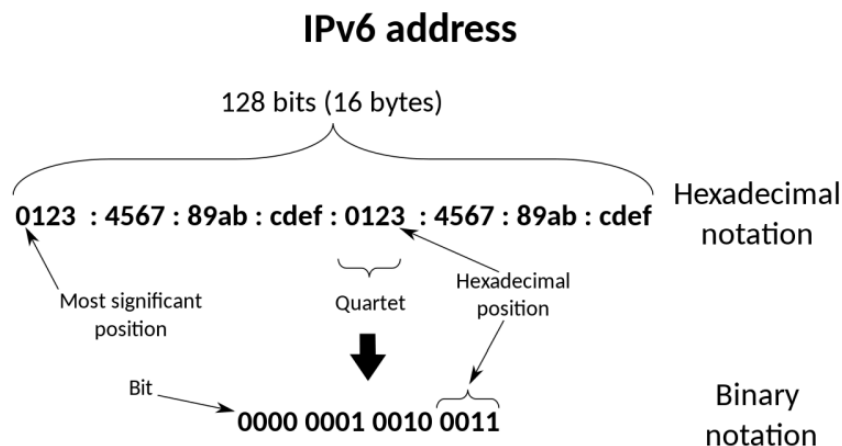
IP Address Classes or Classful addressing

Class in IP v4 Address	Range of Values ONLY in 1 st OCTET	Classful Subnet Mask '/' or CIDR or Prefix Notation	Default Subnet Mask Dotted Decimal
A	1 – 126	/8	255.0.0.0
B	128 – 191	/16	255.255.0.0
C	192 – 223	/24	255.255.255.0
D	224 – 239	Not Applicable	Not Applicable
E	240 – 255	Not Applicable	Not Applicable

- i. **Class A addresses** → designed for large organizations with a large number of hosts or routers. In this the first octet of the address identifies the network and the next three octets are used to identify the host.
- ii. **Class B address** → designed for medium sized organizations with tens of thousands of attached hosts or routers. In this, the first two octets of the address identify the network and the next two octets identify the host within the network.
- iii. **Class C address** → designed for small organizations with a small number of attached hosts or routers. In class C, the first three octets of address are used for network and the last octet is used to identify the host.
- iv. **Class D address** → used for multi-casting. The concept of division of octets into network id and host id does not apply to class D.
- v. **Class E address** → reserved for future use and are experimental. The concept of network id and host-id does not apply on class E also.

IPv6

An IPv6 address is made of 128 bits divided into eight 16-bits blocks. Each block is then converted into 4-digit Hexadecimal numbers separated by colon symbols.

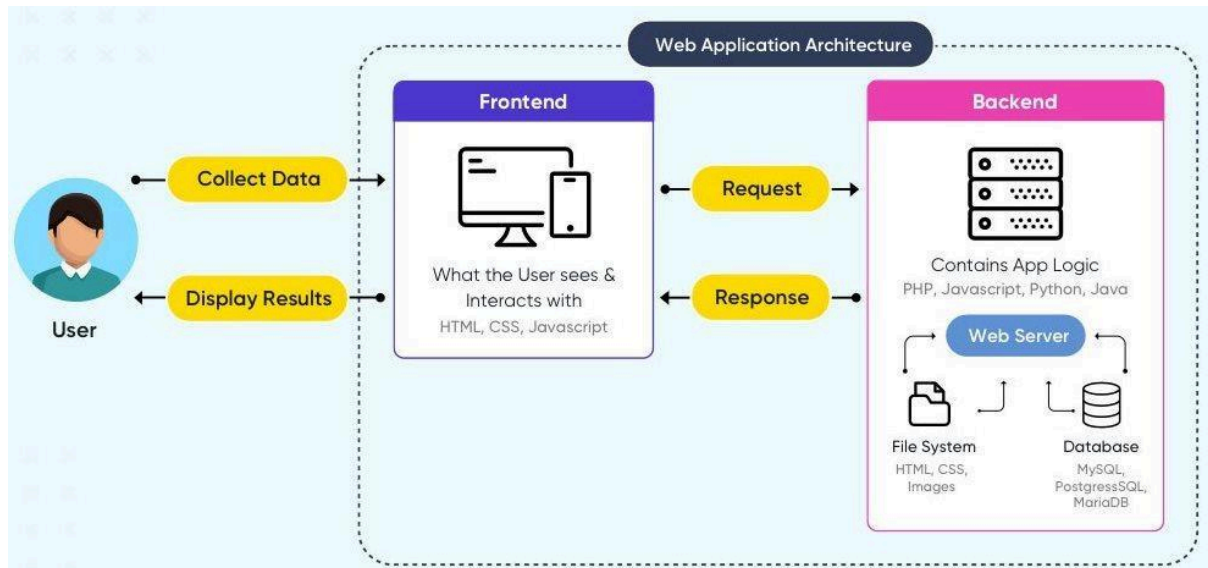


IPv6 vs. IPv4 Comparison

Feature	IPv4	IPv6
Address Size	32-bit	128-bit
Address Format	Decimal	Hexadecimal
Broadcasting	Supported	Replaced by Multicast
Security	Optional	Built-in IPSec

2. Explain in detail about the layout of web application with the software components and define how the data is delivered through HTTP.

Web application architecture is a blueprint of simultaneous interactions between components, databases, middleware systems, user interfaces, and servers in an application.



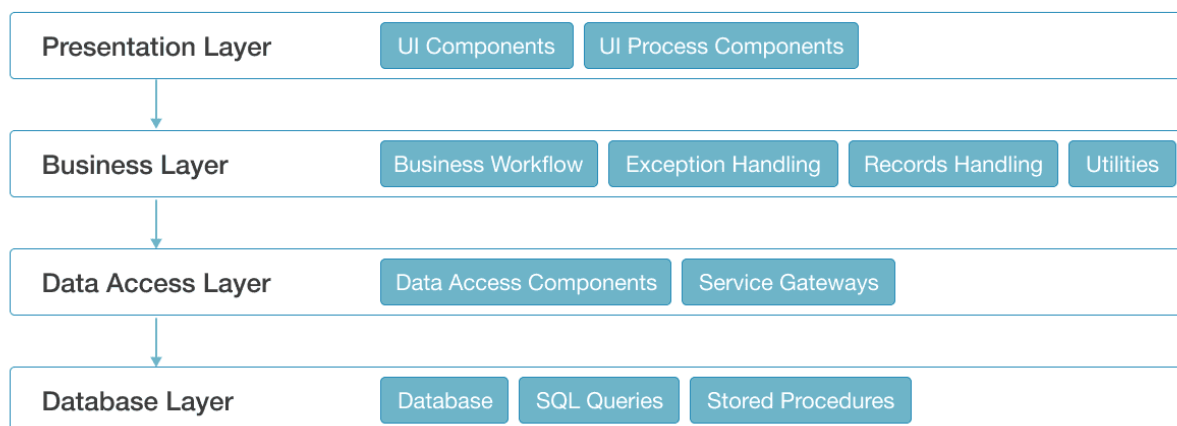
Front End

The frontend of web apps works on the client-side in which the user will interact. The client-side is written in JavaScript, CSS, and HTML.

Back End

The backend is on the server-end that is not accessed by the users. The server is used to store data that can be manipulated easily when required. It can be done easily with platforms like JavaScript, Python, Java, PHP, etc.

A) Layers of Web Application Architecture



i. Presentation Layer

It is built with HTML, CSS, JavaScript, and its frameworks, enabling communication between the interface and browser to facilitate user interaction.

ii. Business Layer

It defines the business logic and rules. It processes browser requests, executes the business logic associated with the requests, and then sends it to the presentation layer.

iii. Data Access Layer

It is responsible for data persistence and is also known as Persistence layer. It's closely connected to the business layer and has a database server that retrieves data from corresponding servers.

iv. Database Layer

It is also known as Data Service Layer, which holds all the data and ensures data security by separating the business logic from the client-side.

B) Components of Web Application Architecture**i. Web Browser**

The browser or the client-side component or the front-end component is the key component that interacts with the user, receives the input and manages the presentation logic.

ii. Web Server

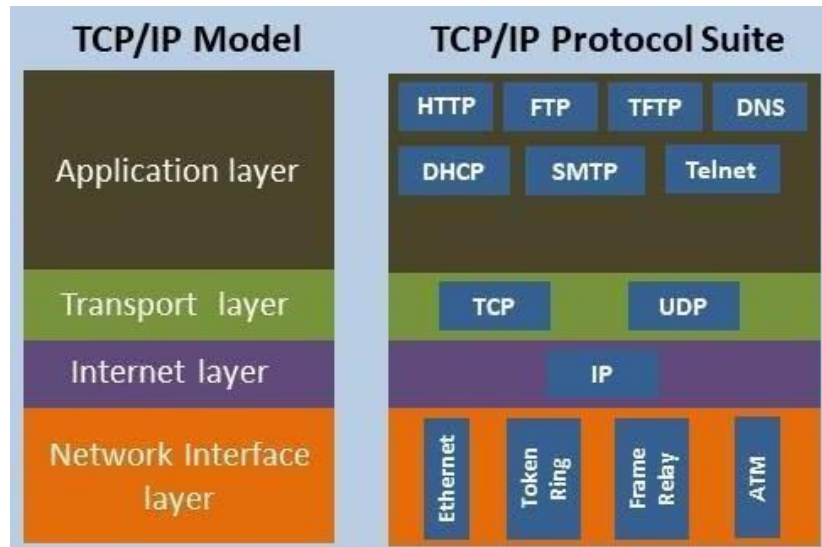
The web server handles the business logic and processes the user requests by routing the requests to the right component and managing the entire application operations.

iii. Database Server

The database server provides the required data for the application.

3. Explain the model that defines how data is transmitted over networks, ensuring reliable communication between devices.

TCP/IP stands for Transmission Control Protocol/ Internet Protocol. TCP/IP Model helps you to determine how a specific computer should be connected to the internet and how data should be transmitted between them. The purpose of the TCP/IP model is to allow communication over large distances. TCP/IP Stack is specifically designed as a model to offer highly reliable and end-to-end byte stream over an unreliable internetwork.



A) Application Layer

Application-layer helps you to identify communication partners, determine resource availability, and synchronize communication. It allows users to log on to a remote host. This layer provides various E-mail services. This application offers distributed database sources and access for global information about various objects and services.

(i) HTTP and HTTPS

HTTP stands for Hypertext transfer protocol. It is used by the World Wide Web to manage communications between web browsers and servers. HTTPS stands for HTTP-Secure. It is a combination of HTTP with SSL(Secure Socket Layer). It is efficient in cases where the browser needs to fill out forms, sign in, authenticate, and carry out bank transactions.

(ii) SSH

SSH stands for Secure Shell. It is similar to Telnet. The reason SSH is preferred is because of its ability to maintain the encrypted connection. It sets up a secure session over a TCP/IP connection.

B) Transport Layer

Transport layer builds on the network layer in order to provide data transport from a source system to a destination system. It determines how much data should be sent where and at what rate. It helps ensure that data units are delivered error-free and in sequence. Transport layer helps you to control the reliability of a link through flow control, error control, and segmentation or de-segmentation.

(i) TCP

Applications can interact with one another using TCP as though they were physically connected by a circuit. TCP transmits data in a way that resembles character-by-character transmission rather than separate packets. A starting point that establishes the connection, the whole transmission in byte order, and an ending point that closes the connection make up this transmission.

(ii) UDP

The datagram delivery service is provided by UDP, the other transport layer protocol. Connections between receiving and sending hosts are not verified by UDP. Applications that transport little amounts of data use UDP rather than TCP because it eliminates the processes of establishing and validating connections.

C) Internet Layer

It is also known as a network layer. The main work of this layer is to send the packets from any network, and any computer still they reach the destination irrespective of the route they take. The Internet layer offers the functional and procedural method for transferring variable length data sequences from one node to another with the help of various networks.

(i) IP

IP stands for Internet Protocol and it is responsible for delivering packets from the source host to the destination host by looking at the IP addresses in the packet headers. IP has 2 versions: IPv4 and IPv6. IPv4 is the one that most websites are using currently.

(ii) ICMP

ICMP stands for Internet Control Message Protocol. It is encapsulated within IP datagrams and is responsible for providing hosts with information about network problems.

(iii) ARP

ARP stands for Address Resolution Protocol. Its job is to find the hardware address of a host from a known IP address. ARP has several types: Reverse ARP, and Inverse ARP.

D) Network Interface Layer

This layer is also called a network access layer. It helps you to define details of how data should be sent using the network. This layer defines how the data should be sent physically through the network. This layer is responsible for the transmission of the data between two devices on the same network.

4. Describe various types of networking devices used in computer networks, explaining their functions, differences, and how they contribute to efficient data communication.

Hardware devices that are used to connect computers, printers, fax machines and other electronic devices to a network are called network devices. These devices transfer data in a fast, secure and correct way over the same or different networks. Network devices may be inter-network or intra-network.

A) NIC Card

NIC is used to physically connect host devices to the network media. A NIC is a printed circuit board that fits into the expansion slot of a bus on a computer motherboard. It can also be a peripheral device. NICs are sometimes called network adapters. Each NIC is identified by a unique code called a Media Access Control (MAC) address. This address is used to control data communication for the host on the network.

B) Repeater

A repeater is a network device that functions at the OSI physical layer. Repeaters regenerate analog or digital signals that are distorted by transmission loss due to attenuation. A repeater does not make an intelligent decision concerning forwarding packets.

C) Hub

The hub works on the physical layer of the OSI model. It is one type of broadcast device which is used to connect the devices in the same network. The hub sends the data in the form of binary bits and it only works in half-duplex. The hub doesn't store any IP addresses or MAC addresses and in the hub, only one device can send data at a time. The types of hubs are,

- Active hub
- Passive Hub
- Intelligent Hub

D) Switch

The switch works on the data link layer of the OSI model. It is one type of multicast device which is used to connect the devices in the same network. The switch sends the data in the form of frames and it only works in full-duplex. The switch stores MAC addresses and in the switch, multiple devices can send data at the same time.

E) Bridge

A bridge is the network device that establishes a connection between various LANs thus forming a larger LAN. This scenario of grouping networks is termed network bridging. The functionality of bridges is at the data link layer of the OSI model and so these network devices are also called Layer 2 switches. The types of bridges are,

- Transparent Bridge
- Source Routing Bridge

F) Router

The router works on the network layer of the OSI model which is used to connect two or more different networks. The router sends the data in the form of packets and it only works in full-duplex. The router stores IP addresses and in the router, multiple devices can send data at the same time. They can also connect to a WAN, which allows them to connect LANs that are separated.

G) Gateway

A gateway is a piece of networking hardware used in telecommunications for telecommunications networks that allows data to flow from one discrete network to another. Gateways communicate using more than one protocol to connect a bunch of networks also termed protocol converters and might function at any layer of the OSI model.

H) Firewall

A firewall is a network device or software for controlling network security and access rules. Firewalls are inserted in connections between secure internal networks and potentially insecure external networks such as the Internet. Firewalls are typically configured to reject access requests from unrecognized sources while allowing actions from recognized ones. The vital role firewalls play in network security grows in parallel with the constant increase in cyber attacks.

I) Access Point

An Access Point (AP) is a network device that extends a wired network by providing Wi-Fi connectivity to wireless devices. It acts as a bridge between the wired LAN and wireless devices, allowing devices like laptops, smartphones, and tablets to connect without needing physical cables.

Functions,

- o Extends network coverage
- o Improves network capacity
- o Connects wired and wireless networks
- o Supports multiple users

J) Modem

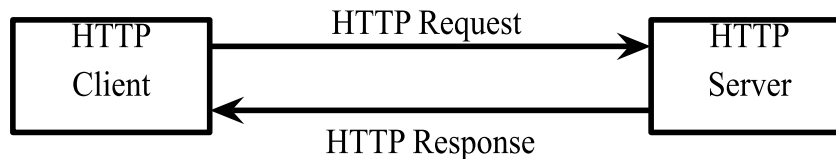
A Modem (Modulator-Demodulator) is a device that converts digital signals from a computer or router into an analog signal for transmission over telephone, fiber, or cable lines. It also converts incoming analog signals back into digital data.

Functions,

- o Converts digital data to analog (modulation) for transmission over ISP lines.
- o Converts analog data back to digital (demodulation) so devices can use it.
- o Connects a network to an ISP (Internet Service Provider).
- o Manages communication speed and data transfer rate.

5. Explain how web browsers and web servers communicate with each other to exchange information and load web pages.

HTTP stands for Hypertext Transfer Protocol. It is the primary protocol used to access the World Wide Web. HTTP is a request-response (also called client-server) protocol that runs over TCP. The common use of HTTP is between a web browser (client) and a web server (server).



A) HTTP Request Headers

A request header is an HTTP header that can be used to provide information about the request context, so that the server can tailor the response.

- Accept: specify desired media type of response
- Accept-Language: specify desired language of response
- Date: date/time at which the message was originated
- Host: host and port number of requested resource

HTTP Request Example

```
GET / HTTP/1.1
Host: www.saveetha.ac.in
HTTP/1.1 200 OK
Content-Type: text/html
```

B) HTTP Response Headers

An HTTP response header is sent by a Web server to a Web browser or client in response to an HTTP request.

- Allow: lists methods supported by request URI
- Content-Language: language
- Content-Type: media type
- Content-Length: length in bytes
- Date: date/time at which the message was originated
- Expires: date/time after which response is considered stale

HTTP Response Example

```

HTTP 1.0 200 OK
Server: NCSA/1.4.2
Content-type: text/html
Content-length: 107
<html>
<body>
The rest of the document goes here
</body>
</html>

```

C) HTTP Status Codes

The status code in a HTTP response is a 3-digit integer where the first digit defines the class of response and the last two digits do not have any categorization role.

Code	Description
1xx: Informational	It means the request has been received and the process is continuing.
2xx: Success	It means the action was successfully received, understood, and accepted.
3xx : Redirection	It means further action must be taken in order to complete the request.
4xx : Client Error	It means the request contains incorrect syntax or cannot be fulfilled.
5xx : Server Error	It means the server failed to fulfill an apparently valid request.

Examples

Code	Description
100	Continue
101	Switching protocols
200	Ok
201	Created
300	Multiple Choices
301	Moved Permanently
400	Bad Request
401	Unauthorized
500	Internal Server Error
501	Not Implemented

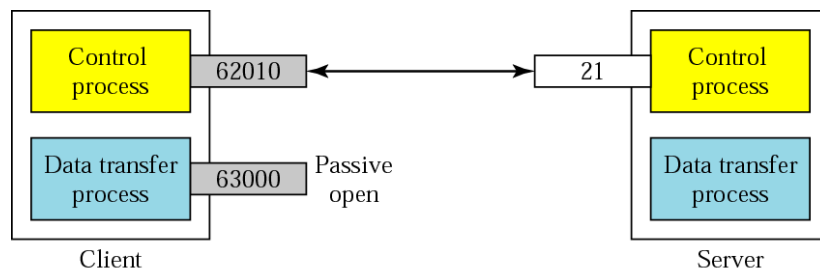
6a. Write short notes on FTP.

The File Transfer Protocol is a standard communication protocol used for the transfer of computer files from a server to a client on a computer network.

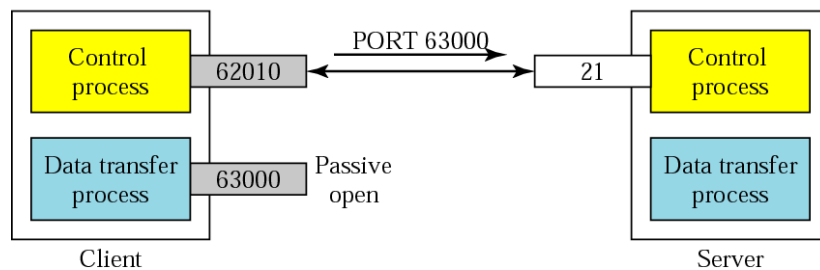
FTP operation

FTP uses client/server model for communication. Two TCP connections are used for file transfer. On one connection control signals are exchanged and the other connection is used for actual data transfer. These two connections are called control connection and data connection.

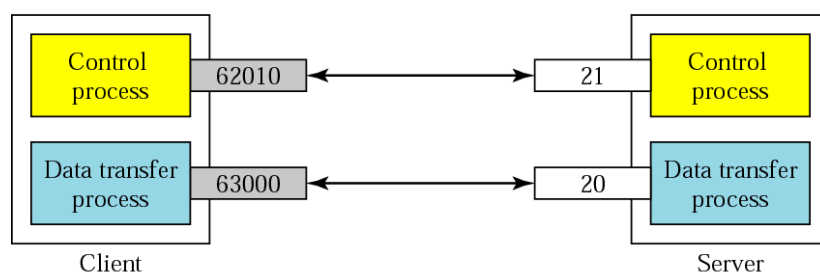
Control and Data Connections



a. Passive open by client



b. Sending ephemeral port number to server



c. Active open by server

Active FTP connection: In an Active FTP connection, the client establishes the command channel and the server establishes the data channel. When the client requests the data over the connection the server initiates the transfer of the data to the client.

Passive FTP connection: In a Passive FTP connection, the client establishes both the data channel as well as the command channel. When the client requests the data over the connection, the server sends a random port number to the client, as soon as the client receives this port number it establishes the data channel.

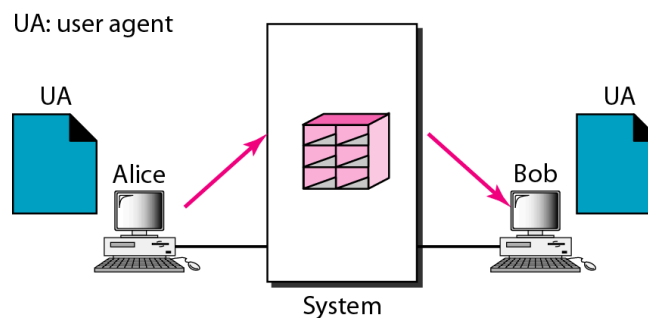
6b. Explain about email.

A mail user agent, which is also known by the name of an email client, is a computer program that gives you the ability to send and retrieve email through the use of the POP and IMAP (Internet Message Access Protocol) protocols through interaction with the user.

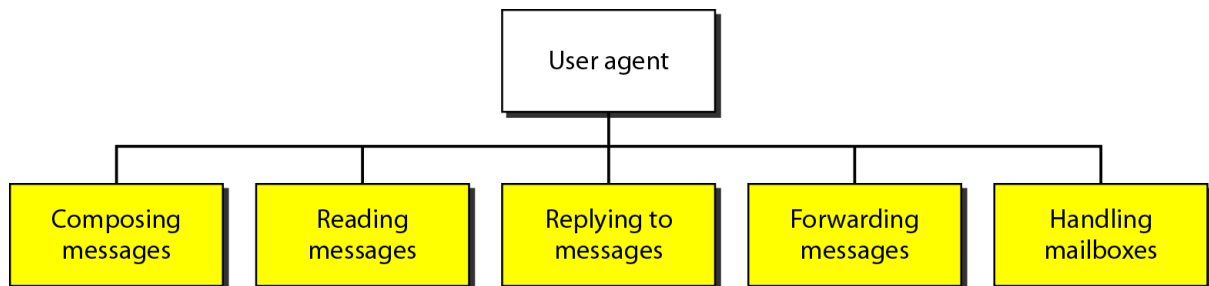
MUAs are the part of the Simple Mail Transfer Protocol (SMTP) system that is responsible for creating email messages and then sending them to a Mail Transfer Agent (MTA). MUAs can be software applications, or they can be webmail services.

MTA (Mail Transfer Agent)

An MTA (Mail Transfer Agent) program or software handles the transfer of e-mail messages between machines using the SMTP protocol. A single email can go through multiple MTAs before reaching the final destination. Users are generally unaware of the existence of these agents, even though each message is sent through at least one MTA.



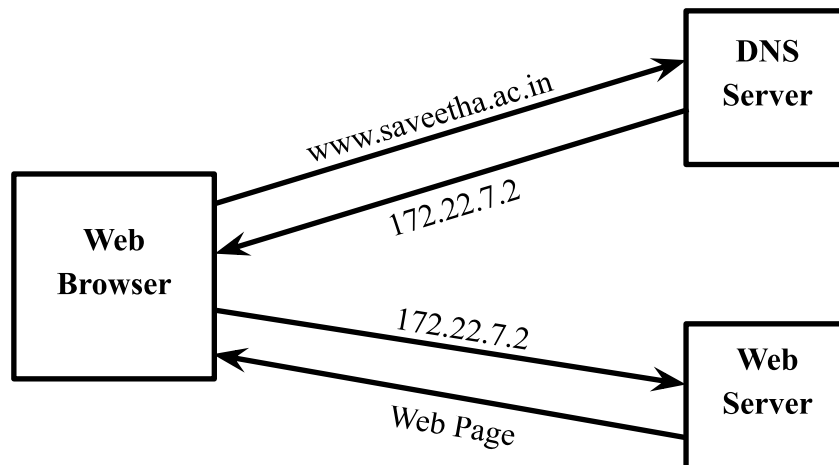
Services of User Agent



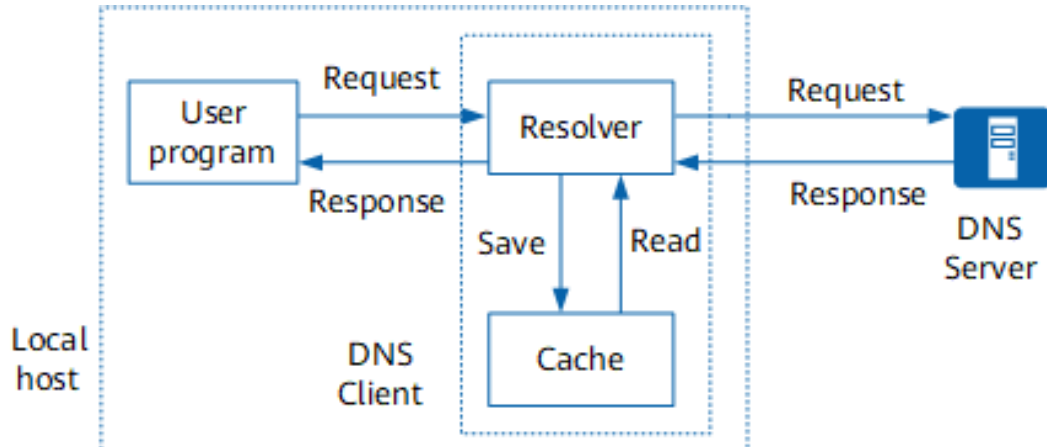
PART C – 15 Marks

1a. Explain the purpose of DNS and how it works?

DNS servers store public IP addresses and other data associated with hostnames. IP addresses are required if you want to access a particular website, device, or network system over the internet. DNS servers are a type of name server.



Working of DNS

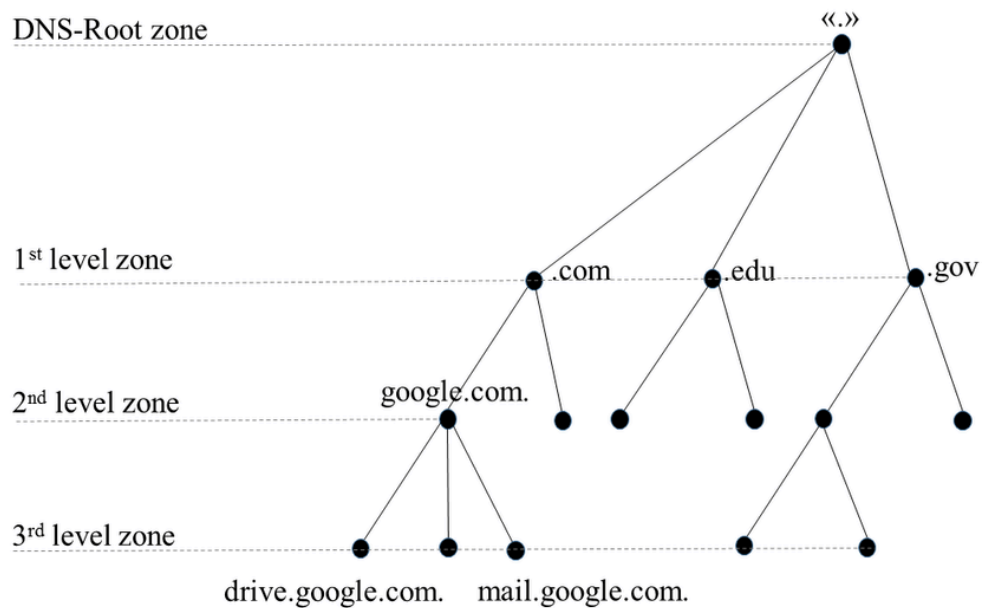


- When a user accesses some applications by domain name, the user program sends a request to the resolver on the DNS client.
- After receiving the request, the resolver searches the local domain name cache.
 - If the domain name matches an entry in the local cache, the resolver sends the corresponding IP address to the user program.
 - If the domain name matches no entry in the local cache, the resolver sends a query message to the DNS server.
- After receiving the response packet from the DNS server, the DNS client sends the resolution result to the user program.

DNS Components

1. Name Space → Specifications for a structured name space and data associated with the names.
2. Resolvers → Client programs that extract information from Name Servers.
3. Name Servers → Server programs which hold information about the structure and the names.

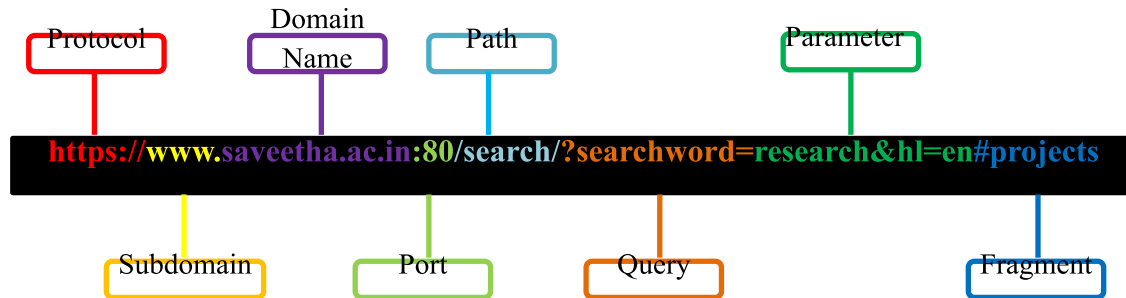
DNS Domains



1b. Describe the various parts of the following URL.

<https://www.saveetha.ac.in:80/search/?searchword=research&hl=en#projects>

URLs, or Uniform Resource Locators, are the schema by which documents or data are addressed in the World Wide Web. URL contains the following information,



Protocol

The protocol declares how your web browser should communicate with a web server when sending or fetching a web page or document.

Subdomain

A subdomain is a subdivision of the main domain name.

Domain name

A domain name is a unique reference that identifies a web site on the internet.

Port

When declared in a URL it comes right after the domain, separated by a colon.

Path

The path typically refers to a file or directory on the web server, e.g. /directory/file.php.

Query

A query is commonly found in the URL of dynamic pages and is represented by a question mark followed by one or more parameters.

Parameters

Parameters are snippets of information found in the query string of a URL.

Fragment

A fragment is an internal page reference, sometimes called a named anchor.

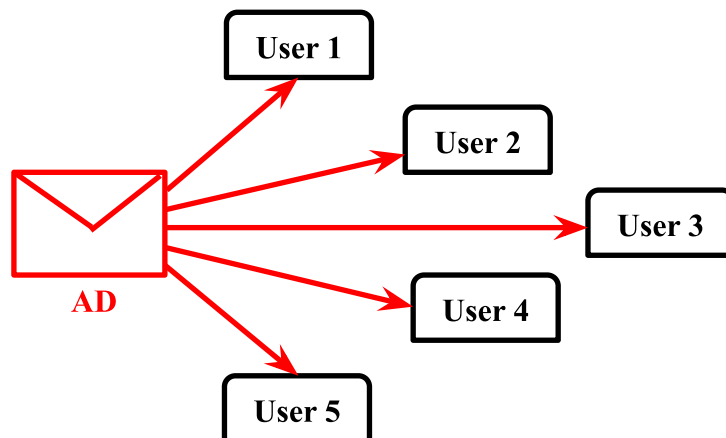
2. Elaborate on the key challenges and risks associated with web security. Discuss common web vulnerabilities, their impact, and possible mitigation strategies.

i. Malicious Websites

A malicious website is a site that attempts to install malware onto your device. Malicious websites, when visited by an unsuspecting victim infect a user machine to steal invaluable information, redirect users to malicious targets.. In the case of a drive-by download, the website will attempt to install software on your computer without asking for permission.

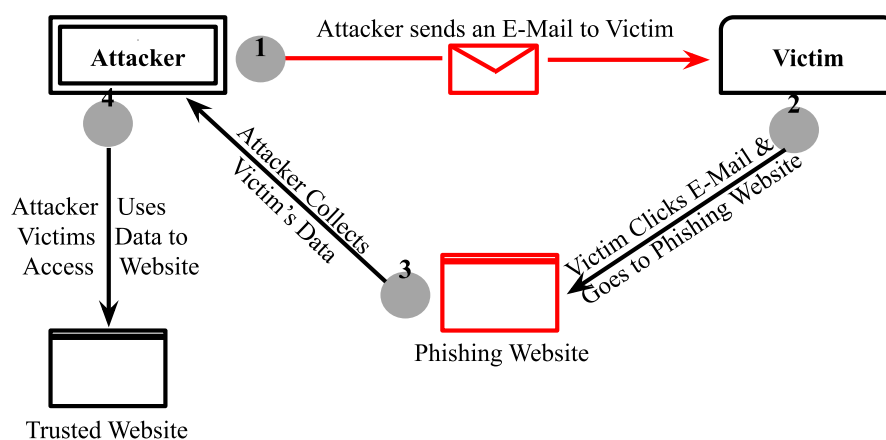
ii. Spam

Spam is defined as irrelevant messages sent to a large number of Internet users, for illegitimate advertising, and other activities such as spreading malware. Spam emails can be dangerous because they may contain malicious links that can infect your computer with malware.



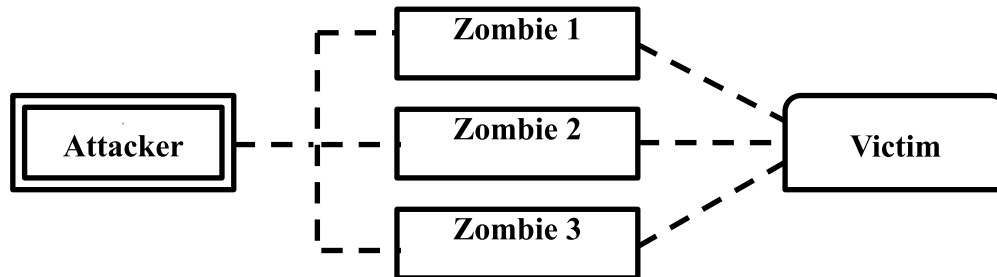
iii. Phishing

This is a method of luring an unsuspecting user into giving out their username and password for a secure web resource. It occurs when an attacker dupes a victim into opening an email, instant message, or text message.



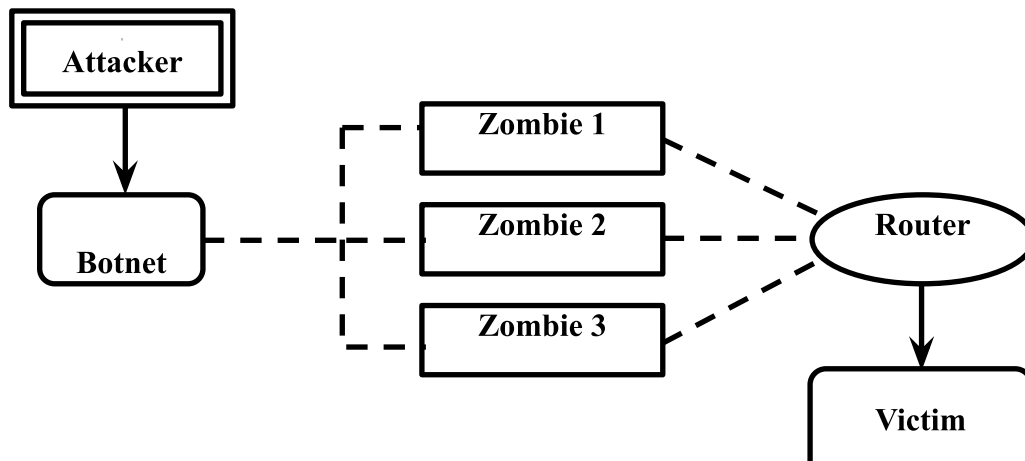
iv. Denial of Service

A Denial-of-Service (DoS) attack is a cyberattack that makes a device or computer inaccessible to its intended users. This is done by overwhelming the target with requests until it can no longer process normal traffic.



v. Distributed DoS Attack

A distributed denial-of-service (DDoS) attack is a cybercrime that floods a target with internet traffic to make it inaccessible. This can be done by using multiple IP addresses or machines to overwhelm the target's resources.



vi. Botnet

A botnet is a collection of compromised computers (called Zombie computers) running programs. They are usually installed via worms, Trojan horses, or backdoors, under a common command and control infrastructure.

vii. SQL Injection

SQL injection (SQLi) is a web security vulnerability that allows an attacker to interfere with the queries that an application makes to its database. This can allow an attacker to view data that they are not normally able to retrieve. This might include data that belongs to other users, or any other data that the application can access. In many cases, an attacker can modify or delete this data, causing persistent changes to the application's content.

viii. Cross Site Scripting

Cross Site Scripting is also shortly known as XSS. XSS vulnerabilities target scripts embedded in a page that are executed on the client side i.e. user browser rather than at the server side.